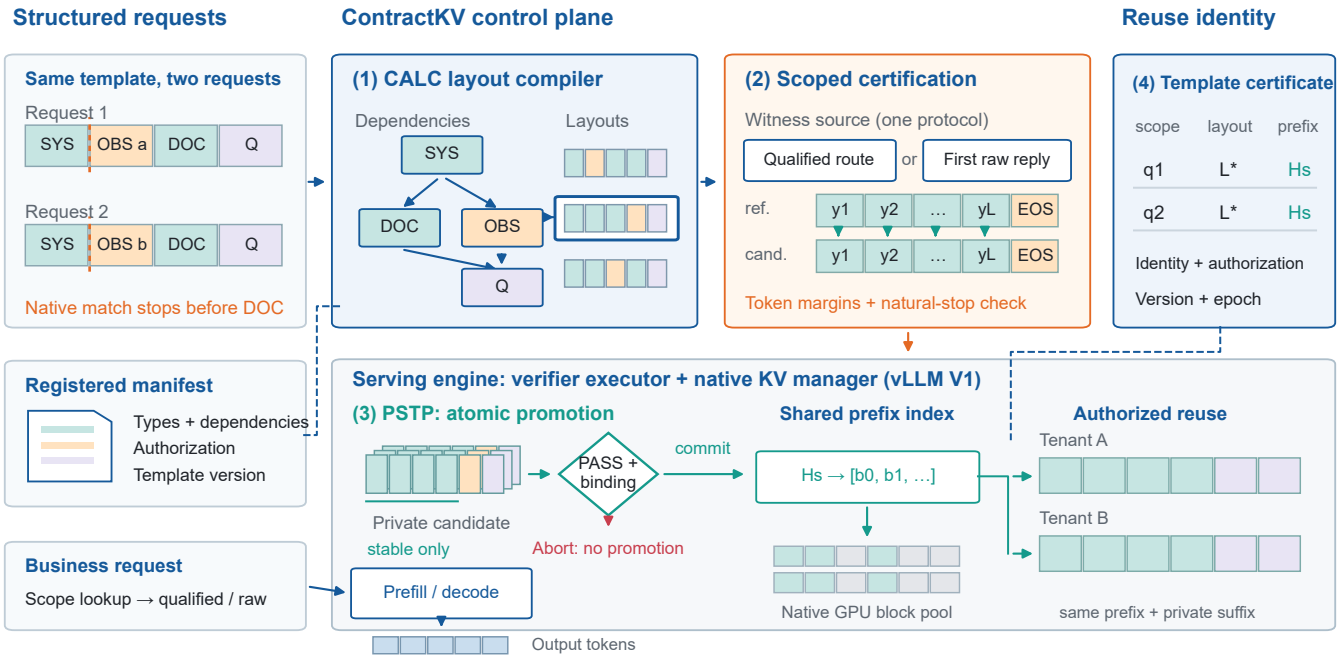
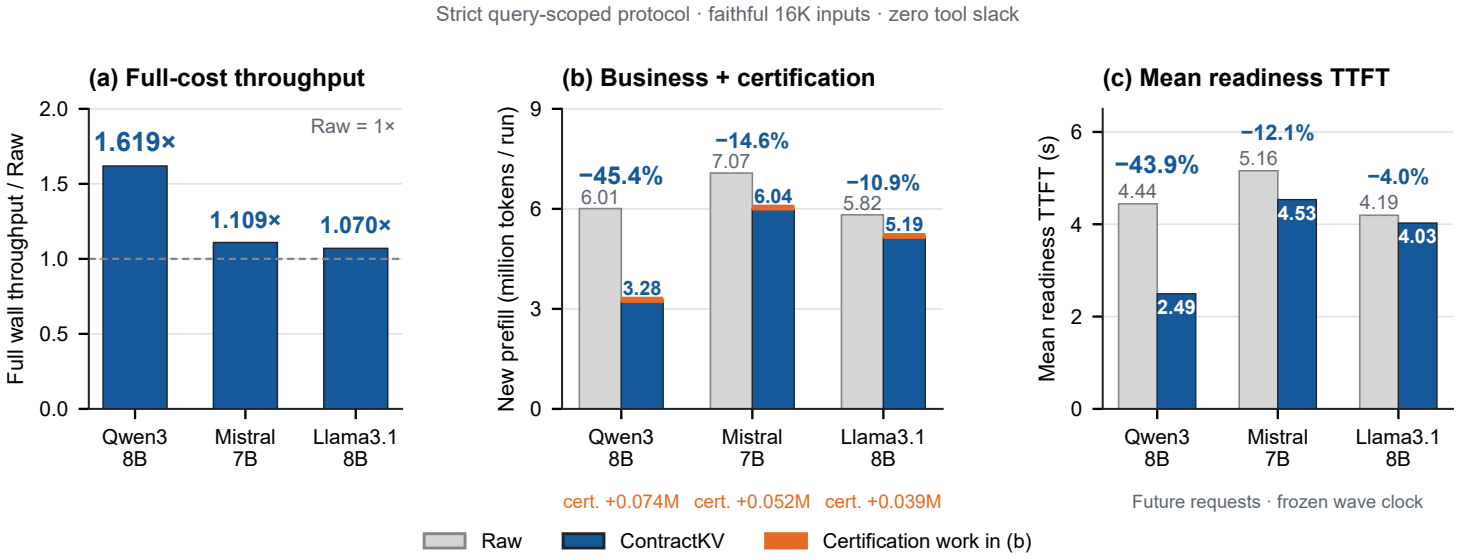
ContractKV | Figure 1

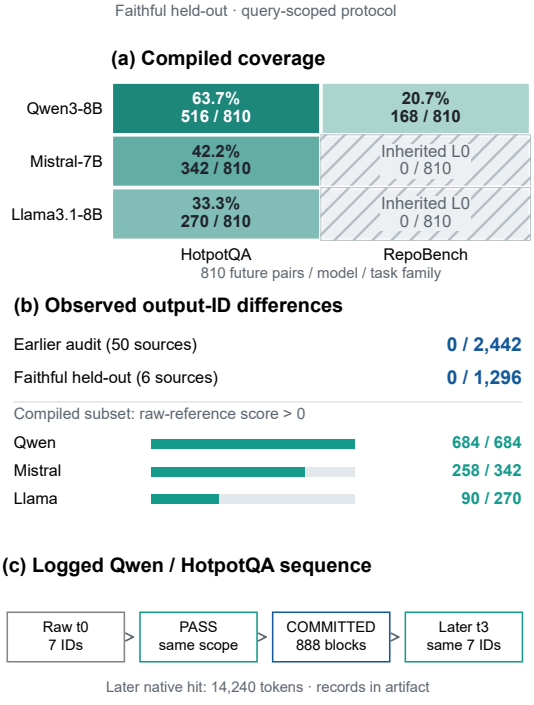


ContractKV makes a compiled stable prefix an evidence-bound service object. CALC repairs a structural prefix gap; one of two separately evaluated witness protocols supplies scoped token-path evidence; PSTP commits eligible state only after PASS and a valid binding. Authorized tenants of the same template share the prefix and keep changing suffixes private. Qualification and physical residency have separate lifetimes.



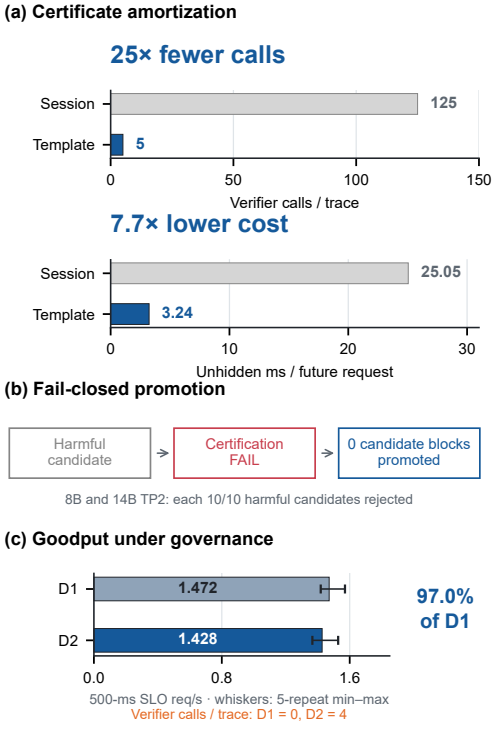
Strict query-qualified performance on six source-disjoint HotpotQA/RepoBench templates, three processes/model, faithful 16K inputs, and zero slack. (a) Throughput charges all business responses, qualification, failed attempts, and synchronous waiting. (b) Actual business plus qualification new-prefill counts per run; orange caps mark certification work at its true scale, with exact added counts below. (c) Mean readiness TTFT over future requests uses the frozen two-request wave clock. Three-repeat values and tail statistics are reported separately.

ContractKV | Figure 3



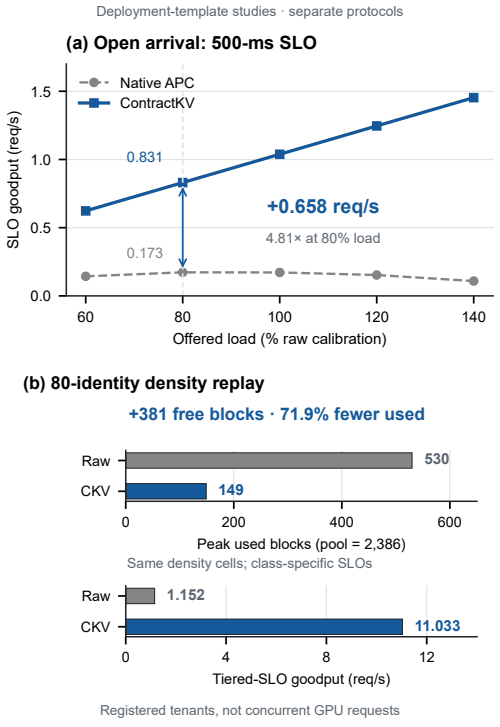
Qualification coverage and free-running evidence. (a) Each model/task cell has 810 future pairs; hatched cells inherit L0. (b) Output-ID differences and raw-reference positive-score counts within compiled subsets. Positive credit is not complete correctness. (c) One logged Qwen/HotpotQA witness, PASS, committed prefix and later identical output; full records are archived.

ContractKV | Figure 4

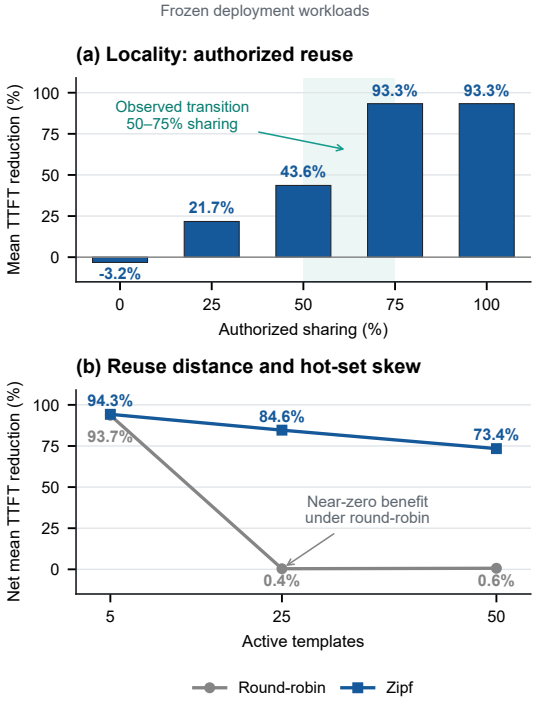


Separate deployment controls. (a) Registration granularity with fixed layout/locality; three processes and 1,125 future requests/trace. (b) Each 8B/14B TP2 probe rejects 10/10 harmful candidates with zero rejected-candidate promotion. (c) Same-engine D1/D2 500-ms SLO goodput; whiskers are five-process min-max, not confidence intervals.

ContractKV | Figure 5



Distinct deployment protocols. (a) Open-arrival 500-ms SLO goodput; the 4.81x ratio is the 80%-load point. (b) The same 80-identity density cells supply both allocator occupancy and goodput under class-specific 750/1,500/3,000-ms SLOs, with a 2,386-block pool. Registered identities are not simultaneous GPU requests.



Operating region in frozen deployment workloads. (a) The shaded 50-75% sharing interval is an observed transition, not a universal threshold. (b) Access-pattern sensitivity to reuse distance and hot-set skew. Both axes report TTFT reduction; neither is a direct residency fraction.